



Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I - III Semester(2025-26)
Faculty Name	Ms. Alluri Swathi
Student Name	Sivarapu Rishil
Roll Number	2520090174
Branch	CSIT
Course Outcome	4

Case Study Topic: UrbanAI – Intelligent Smart City Infrastructure Optimization Platform Using Dynamic Programming

Work Done Summary:

Implemented 0/1 Knapsack Dynamic Programming in Java to optimize resource allocation for smart city infrastructure projects. Given a limited budget (knapsack capacity) and a set of potential infrastructure projects (e.g., smart traffic lights, EV charging stations, air quality sensors), each with a cost and an estimated urban benefit value, the algorithm selects the optimal combination of projects to maximize total benefit without exceeding the budget. Compared the DP approach with a greedy strategy and demonstrated that DP guarantees an optimal solution for resource-constrained urban planning problems.

Implementation Details :

- *Created a class SmartCityProject with attributes: project name, cost, benefit value.
- *Implemented knapsackDP() using a 2D table to store maximum benefit for subproblems.
- *Used backtracking to recover the selected projects.
- *displaySelectedProjects() shows the optimal set of infrastructure investments.
- *The main method initializes a list of candidate projects, defines a budget, computes the optimal selection, and prints results with total benefit and cost.
- *Compared DP output with a greedy “benefit-per-cost” selection to highlight DP’s optimality.

Result : Screenshots/Output:

```
=== UrbanAI Smart City Investment Optimization (0/1 Knapsack DP) ===  
Optimal Total Benefit: 115  
Selected Projects (maximizing benefit within budget $15M):  
- Public Wi-Fi Hotspots | Cost: $6M | Benefit: 50  
- Smart Street Lighting | Cost: $4M | Benefit: 40  
- Smart Traffic Lights | Cost: $5M | Benefit: 45  
Total Cost: $15M
```

Time Complexity:

- *DP Table Initialization: $O(n \times \text{budget})$ where n = number of projects, budget = capacity (integer)
- *DP Table Filling: $O(n \times \text{budget})$ – each cell computed in constant time
- *Backtracking to find selected projects: $O(n)$
- *Overall Complexity for Knapsack DP: $O(n \times \text{budget})$ – pseudo-polynomial, efficient for moderate budget values in city planning
- *Space Complexity: $O(n \times \text{budget})$ for the DP table (can be optimized to $O(\text{budget})$ using 1D array)

GitHub Link : https://github.com/rishill10/DSA2_CO4

Student Signature	Faculty Signature

